

Table 1: Performance and computational efficiency of Qwen3-8B under the sequential configuration. Acc. ($\uparrow$) denotes task accuracy (%), or pass@1 (%) for code generation; Token ($\downarrow$) denotes tokens per problem; and Time ($\downarrow$) denotes inference time (seconds per problem). Single, Text, and Linear Align times are converted to the Kernel machine’s throughput. AVG denotes the unweighted mean across all six benchmarks. The highest accuracy among training-free latent alignment methods, the lowest token count, and the shortest inference time in each column (including AVG) are shown in bold.

Method	Metric	AIME 2024	AIME 2025	Human Eval+	MBPP+	GPQA- Diamond	MedQA	AVG
Single	Acc.	50.00	46.70	74.40	64.80	39.90	53.00	54.80
	Token	12891	14692	2507	2053	6435	2098	6779.33
	Time	340.2	377.8	31.5	28.9	104.6	18.3	150.22
Text	Acc.	53.30	53.30	80.50	69.50	43.40	75.00	62.50
	Token	38596	45088	4593	3695	17986	4260	19036.33
	Time	1018.6	1159.4	57.7	52.1	292.2	37.1	436.18
Linear Align (?)	Acc.	56.70	53.30	80.50	74.60	45.50	75.30	64.32
	Token	8953	8699	1866	1164	4571	1555	4468.00
	Time	236.3	223.7	23.4	16.4	74.3	13.5	97.93
Exact-Soft Align (?)	Acc.	63.75	49.17	82.62	73.15	49.24	66.67	64.10
	Token	22342	24557	15055	14388	13076	15972	17565.00
	Time	549.0	597.0	112.4	101.6	257.6	111.1	288.12
Kernel Align	Acc.	66.25	49.58	83.54	75.26	48.74	73.67	66.17
	Token	8548	10158	1637	1462	3588	1139	4422.00
	Time	225.6	261.2	18.8	17.9	58.3	12.5	99.05
Kernel-ES Align	Acc.	64.17	50.42	83.38	75.20	48.61	72.00	65.63
	Token	8770	9830	1696	1528	3846	1154	4470.67
	Time	225.8	255.5	21.5	19.6	59.4	13.6	99.23

Table 2: Performance and computational efficiency of Qwen3-8B under the hierarchical configuration. Acc. ($\uparrow$) denotes task accuracy (%), or pass@1 (%) for code generation; Token ($\downarrow$) denotes tokens per problem; and Time ($\downarrow$) denotes inference time (seconds per problem). Single, Text, and Linear Align times are converted to the Kernel machine’s throughput. AVG denotes the unweighted mean across all six benchmarks. The highest accuracy among training-free latent alignment methods, the lowest token count, and the shortest inference time in each column (including AVG) are shown in bold. A dash indicates an unavailable result; AVG is reported only when all six benchmark results are available.

Method	Metric	AIME 2024	AIME 2025	Human Eval+	MBPP+	GPQA- Diamond	MedQA	AVG
Single	Acc.	50.00	46.70	74.40	64.80	39.90	53.00	54.80
	Token	12891	14692	2507	2053	6435	2098	6779.33
	Time	335.6	379.0	30.7	33.8	102.7	25.1	151.15
Text	Acc.	53.30	50.00	76.80	71.90	43.00	76.30	61.88
	Token	42629	53929	8768	7703	22450	6893	23728.67
	Time	1109.7	1391.2	107.5	126.8	358.4	82.5	529.35
Linear Align (?)	Acc.	53.30	50.00	78.00	72.20	46.90	77.00	62.90
	Token	7526	13230	1274	1264	3395	1007	4616.00
	Time	195.9	341.3	15.6	20.8	54.2	12.1	106.65
Exact-Soft Align (?)	Acc.	–	–	–	73.21	–	–	–
	Token	–	–	–	12768	–	–	–
	Time	–	–	–	92.2	–	–	–
Kernel Align	Acc.	60.00	53.33	83.84	72.75	51.39	73.33	65.77
	Token	9926	11207	1681	1502	3752	1351	4903.17
	Time	258.4	289.1	20.1	20.1	59.9	15.7	110.55
Kernel-ES Align	Acc.	65.83	52.50	83.23	72.16	49.75	72.67	66.02
	Token	9898	11300	1637	1460	3964	1336	4932.50
	Time	257.9	291.9	21.0	20.2	61.5	14.8	111.22

Table 3: Performance and computational efficiency of Qwen3-14B under the sequential configuration. Acc. ($\uparrow$) denotes task accuracy (%), or pass@1 (%) for code generation; Token ($\downarrow$) denotes tokens per problem; and Time ($\downarrow$) denotes inference time (seconds per problem). Single, Text, and Linear Align times are converted to the Kernel machine’s throughput. AVG denotes the unweighted mean across all six benchmarks. The highest accuracy among training-free latent alignment methods, the lowest token count, and the shortest inference time in each column (including AVG) are shown in bold. A dash indicates an unavailable result; AVG is reported only when all six benchmark results are available.

Method	Metric	AIME 2024	AIME 2025	Human Eval+	MBPP+	GPQA- Diamond	MedQA	AVG
Single	Acc.	63.30	56.70	76.80	68.50	48.50	64.70	63.08
	Token	11263	11298	2366	1858	5547	1746	5679.67
	Time	348.6	369.2	31.8	28.5	109.5	21.1	151.45
Text	Acc.	63.30	60.00	81.10	72.80	51.50	80.30	68.17
	Token	32092	44618	5934	4971	12676	3444	17289.17
	Time	993.2	1458.1	79.7	76.2	250.2	41.6	483.17
Linear Align (?)	Acc.	66.70	63.30	86.50	75.70	52.00	80.70	70.82
	Token	10593	11402	2042	1621	5454	1841	5492.17
	Time	327.8	372.6	27.4	24.9	107.7	22.2	147.10
Exact-Soft Align (?)	Acc.	–	49.58	90.70	–	53.54	79.00	–
	Token	–	21403	19587	–	18528	22875	–
	Time	–	575.5	157.3	–	222.9	151.0	–
Kernel Align	Acc.	72.08	56.67	88.11	78.11	55.81	81.67	72.08
	Token	8259	10266	1594	1336	3009	922	4231.00
	Time	255.6	335.5	24.1	21.5	59.4	13.2	118.22
Kernel-ES Align	Acc.	69.58	60.00	86.43	79.43	54.42	79.33	71.53
	Token	8489	10402	1607	1346	2995	912	4291.83
	Time	283.4	350.6	28.1	25.5	67.2	14.3	128.18

Table 4: Performance and computational efficiency of Qwen3-14B under the hierarchical configuration. Acc. ($\uparrow$) denotes task accuracy (%), or pass@1 (%) for code generation; Token ($\downarrow$) denotes tokens per problem; and Time ($\downarrow$) denotes inference time (seconds per problem). Single, Text, and Linear Align times are converted to the Kernel machine’s throughput. AVG denotes the unweighted mean across all six benchmarks. The highest accuracy among training-free latent alignment methods, the lowest token count, and the shortest inference time in each column (including AVG) are shown in bold. A dash indicates an unavailable result; AVG is reported only when all six benchmark results are available.

Method	Metric	AIME 2024	AIME 2025	Human Eval+	MBPP+	GPQA- Diamond	MedQA	AVG
Single	Acc.	63.30	56.70	76.80	68.50	48.50	64.70	63.08
	Token	11263	11298	2366	1858	5547	1746	5679.67
	Time	357.6	364.4	26.4	16.3	123.2	31.6	153.25
Text	Acc.	70.00	66.70	84.10	73.00	52.00	78.00	70.63
	Token	29025	50003	8114	7458	20931	5473	20167.33
	Time	921.7	1612.8	90.6	65.4	464.9	99.0	542.40
Linear Align (?)	Acc.	73.30	66.70	86.60	73.80	53.00	78.30	71.95
	Token	10230	9527	1512	1187	3606	899	4493.50
	Time	324.8	307.3	16.9	10.4	80.1	16.3	125.97
Exact-Soft Align (?)	Acc.	71.25	–	91.16	–	54.55	48.33	–
	Token	23924	–	21403	–	20909	23700	–
	Time	639.9	–	185.6	–	287.8	173.7	–
Kernel Align	Acc.	69.17	59.17	87.04	78.04	54.80	79.00	71.20
	Token	8947	10755	1454	1132	2769	860	4319.50
	Time	284.1	346.9	24.0	21.6	61.5	12.4	125.08
Kernel-ES Align	Acc.	69.58	57.92	86.74	78.31	55.81	77.67	71.01
	Token	9017	11122	1430	1182	2755	875	4396.83
	Time	291.0	365.7	25.4	23.7	63.6	13.8	130.53

Table 5: Performance and computational efficiency of cross-model communication between Qwen3-8B and Qwen3-14B. Arrows indicate the sender-to-receiver direction. Acc. ($\uparrow$) denotes task accuracy (%), or pass@1 (%) for code generation; Dec. ($\downarrow$) denotes the number of explicit vocabulary-decoding operations per problem; and Time ($\downarrow$) denotes end-to-end inference time (seconds per problem). Receiver-only answers the original question without sender representations, using the same receiver generation settings. AVG denotes the unweighted mean across all nine benchmarks. A dash indicates an unavailable result.

Method	Metric	AIME 2024	AIME 2025	Human Eval+	MBPP+	ARC-C	GPQA- Diamond	MedQA	AVG
<i>Qwen3-14B $\rightarrow$ Qwen3-8B</i>									
Linear Align	Acc.	–	–	–	–	–	–	–	–
	Dec.	–	–	–	–	–	–	–	–
	Time	–	–	–	–	–	–	–	–
Exact-Soft Align	Acc.	–	–	–	–	–	–	–	–
	Dec.	–	–	–	–	–	–	–	–
	Time	–	–	–	–	–	–	–	–
Kernel Align	Acc.	–	–	–	–	–	–	–	–
	Dec.	–	–	–	–	–	–	–	–
	Time	–	–	–	–	–	–	–	–
<i>Qwen3-8B $\rightarrow$ Qwen3-14B</i>									
Linear Align	Acc.	–	–	–	–	–	–	–	–
	Dec.	–	–	–	–	–	–	–	–
	Time	–	–	–	–	–	–	–	–
Exact-Soft Align	Acc.	–	–	–	–	–	–	–	–
	Dec.	–	–	–	–	–	–	–	–
	Time	–	–	–	–	–	–	–	–
Kernel Align	Acc.	–	–	–	–	–	–	–	–
	Dec.	–	–	–	–	–	–	–	–
	Time	–	–	–	–	–	–	–	–